Nestify user guide

Version 1.0.0 · 21 October 2026 · Elton Sibara
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Start a new house

The first time Nestify opens, a short welcome shows the four steps (get a plan, shape it, check it, hand it over). Open it again from Help (?) → Welcome tour.

Press **New house** (the house-plus icon, top left):

- **Describe it** in your own words, for example "3 bedrooms, 2 bathrooms with en-suite, open plan, double garage and a study, about 180 m2" or "double storey, 4 bedrooms, laundry". The line underneath shows what Nestify understood as you type. Press **Build it** and you get a full house: walls, passage, doors, windows sized for daylight, room names, a garage with driveway and car, stairs for two storeys, and furniture, lights, sockets and plumbing in every room. Every generated plan passes the building rules check.
- **Starter plans**: Starter home, Family home, Large family home, Double storey.
- **Blank page** to draw your own with the wall, rectangle and line tools.

Everything it makes is ordinary walls and objects, so you edit it like any other plan.

Draw the plan in 2D

The pencil-and-ruler button (top bar, and on the left rail) opens the **2D plan board**: a flat drawing

board. The plan is **its own drawing**, saved with the project but separate from the 3D house: what you built in 3D does not show on it, and drawing on it does not change the 3D house. When the plan is ready, **Build 3D** (top) puts that floor's walls, doors, windows, room names and furniture into the 3D house (boxes and shapes built in 3D stay; Ctrl+Z in 3D undoes it). **Start from the 3D walls** copies the 3D house's walls into an empty plan. "+ Floor" adds a floor to the plan.

- **Tools** (left): Select / move (L, D, W, O ... keys too), **Wall** (click corner after corner, type a length + Enter for an exact wall, click the first corner to close a room, Esc or right-click ends; it snaps to corners, to walls and to straight lines), **Door** and **Window** (click on a wall; choose the kind on the right), **Opening**, **Room name**, **Measure**, **Text**, **Furniture** (the library; the piece lands in the middle of the view, drag it into place), **Erase** and the **Hand**.
- **Select**: drag furniture, slide doors and windows along their wall, drag a wall, or drag a wall's round end to move that corner (the walls joined to it follow). The panel on the right changes a wall's length, thickness and height, a door or window's width, turns furniture, names rooms. Delete removes what is selected; Ctrl+Z undoes.
- **Looks** (top): **Colour** (red walls, tiles, furniture from above), **Technical** (black walls, dimension chains, sizes), **Blueprint** and **Sketch**. Switch any time; Nestify remembers your choice.
- **Save**: this view as a picture in the look you chose, the colour plan, or the A3 plans to print.
- The mouse wheel zooms where the mouse is; right-drag or middle-drag slides the paper; Fit shows the whole floor.

Doors and windows: select one and drag its round ends to make it wider or narrower, or type its width, height and (windows) sill on the right, use - 100 / + 100 mm, or pick a standard size. **Split (S)** cuts a wall in two where you click (each half can then be moved or changed), or cuts a railing in two.

Fixing railings, roofs and pergolas: select one and its corners show as round dots: drag a dot to move that corner, click a small + in the middle of a side to add a corner, click a corner and press Delete to remove it. **Split** on a roof (or pergola) cuts it in two with two clicks (where the cut starts and ends); select a piece and Delete removes that part. **Review** (top bar) shows a quick 3D sketch of the plan and a list of checks (walls whose ends do not meet, rooms without names, missing doors or windows, doors taller than their walls, roofs not over walls); Build 3D shows the same review first, with **Back to fix**.

Import a plan

Press the plan icon (top left), or drop a file on the window. Pictures (JPG, PNG, BMP, TIFF, WebP) and PDFs (every page; use the arrows to pick the one with the floor plan) are accepted.

1. **Sheet** – in a photo the paper is found and flattened automatically (perspective removed, standard paper shape kept), then squared up. If the guess is off, go to step 1 and drag the 4 cyan corners. A scan or PDF page is used as it is.
2. **Walls** – every wall is found: solid, hatched or hollow (drawn as two lines), straight or slanted. The sheet frame, title block, dimension lines, text, furniture, stairs and door swings are ignored, and the walls are joined at corners and T-junctions. Drag an end to fix it (ends that meet move together), drag on empty paper to add a missing wall, tap a wall to delete or split it.
3. **Size** – tap one wall you know (the longest is best), type its real length and press Set (or drag the ruler over any known length). Set the wall height; thickness is measured from the drawing.

4. **Build 3D** – gaps in walls become doors and windows drawn in solid walls become windows.

Moving round the view

Drag turns the view. To **slide** it (the ground and the house move together, as if you dragged the paper): right-drag, Shift-drag, the arrow keys, or turn on the **hand** (H, or the hand button on the left rail) and drag. The wheel zooms, F frames the house, T looks straight down.

Editing walls, doors and windows

Tool	Key	What it does
Select	Space	Click a wall, the floor or the roof. Drag a selected wall's **grip** to move that corner – walls joined there follow.
Wall	L	Click corner after corner. Snaps to **Endpoint**, **Midpoint**, **On wall**, **Intersection**, the **red** / **green axes**, **Parallel** / **Perpendicular** and **From point**. Type a length + Enter for an exact wall. Ending on a wall splits it, so the joint stays editable.
Move	M	Drag a wall; walls joined to it stretch (Ctrl pulls it free). Drag a door/window along its wall. Type a distance + Enter afterwards for an exact move.
Rotate	Q	Drag to turn the selected wall (snaps to 15°).
Push/ Pull	P	Drag any face: the top changes height, a side changes thickness (walls) or width/ depth (objects), a wall end changes length. The face lights up orange; the opposite face stays put (Ctrl: both sides). Type a size + Enter, or + / – for 5 cm steps.
Split	S	Click a wall where it should become two.
Opening	D	Drag along a wall, then pick a door, window, passage or separate wall.

Click the orange arrow strip on the Layers or Properties panel (or press `[` / `]`) to fold it away for a bigger 3D view; on a small window they fold away by themselves. T toggles top view, F frames the house, Delete removes, Esc cancels, Ctrl+Z / Ctrl+Y undo / redo, Ctrl+S save, Ctrl+O open. The **Measurements** box (bottom right) accepts `3.5`, `3500mm`, `350cm`, `12ft`, `30in`.

Keyboard resizing of the selection: Ctrl+←/→ thinner/thicker (narrower/wider), Ctrl+↑/↓ lower/higher, Alt+←/→ shorter/longer (shallower/deeper), Shift for 1 cm steps. A tips card in the 3D view (lightbulb button) always lists the keys for what is selected.

Drawing shapes

- **Rectangle (R)** and **Circle (C)**: draw on the ground or on the top face of anything, then move the mouse up and click to pull the shape into a solid (or type `3,2` / a radius, Enter, then a height, Enter). Draw on top of a solid to stack another one. Solids are objects: Move, Rotate, Push/Pull, copy, delete.
- **Line (N)**: click corner after corner for any outline, **A** makes the next side an arc (click its end, move to bend it, click); close on the first corner or Enter, then pull it up into a solid.

- **Paint bucket (B)**: choose a material or a colour, click walls, roof, floors, rooms or solids.
- **Groups**: Ctrl+click to select several things, **Ctrl+G** groups them (move, turn, copy and delete together), **Ctrl+Shift+G** ungroups.
- **Scenes**: save the current view (camera, style, floor) and glide back to it.
- **Tags**: show or hide whole kinds of things (Furniture, Electrical, Plumbing, Garden, Shapes...).

Boxes as houses: doors, windows, split, paint one face, walk through

Many people build with the Rectangle tool: draw, pull up into a box, and use the boxes as walls.

- **Door, window, passage** (left rail): drag across the side of a box where the opening goes. The card offers the usual doors and windows (or a plain passage); the hole is cut right through and a door leaf or pane of glass is put in it.
- **Split** cuts a box in two where you click, so each part can be moved, painted or pulled on its own.
- **Paint bucket** on a box paints only the face you click; Shift + click paints the whole box. To paint part of a face, draw round it with Dots and choose **Paint area**.
- **Walk** works in houses made of boxes too: their sides are walls you cannot walk through, and their doors stand open so you can walk in.
- **A box inside a bigger one**: click again at the same place to pick the next thing behind (each click goes one deeper, then back to the front). Clicking through an opening picks what is behind it.

Dots push/pull (shape anything)

The Dots tool (left rail, or **U**) shapes the solids you draw - like SketchUp's draw-on-a-face and Push/Pull.

1. Pick the Dots tool. A card at the bottom of the 3D view asks how to join the dots: **Straight lines** or **Smooth curve** (Nestify remembers your choice).
2. Click a face of a shape (a box from Rectangle, a cylinder from Circle, a Line outline): that is dot 1.
3. Click more dots around the outline you want. They snap to corners and edges. Backspace takes one back.
4. Click the first dot again (or double-click, or Enter). The outline turns **yellow** and a card asks what next: * **Edit dots** (or **E**): drag any dot to a new place, click on a line to add a dot there, Backspace removes the last one. **Done** (or Enter) when the outline looks right, like editing a path in Photoshop. * **Paint area**: asks what to paint with (a colour, water, tiles, grass ...) and paints only the inside of the outline, for example a darker panel on a wall or a doormat on a floor. * **Make a pool** (on a top face): digs the outline 1.4 m into the shape and fills it with clear water just below the top: a swimming pool or a pond of any shape. Select "Pool water" to change its level or colour. * **Push** / **Pull** (or Enter, or just right-drag): the outline turns **green**, ready to push or pull.

5. Hold the **right mouse button** and drag: **into** the face pushes the outline in (a recess, a window bay, a hole right through if you go far enough), **out** of the face pulls it out (a bump, a bay, a balcony box). The shape changes as you drag; let go to keep it. Looking straight at a face, drag up to pull and down to push. Or type a distance + Enter (negative pushes in).

If your dots go round (nearly) the whole face, the whole face moves, like Push/Pull. Every new face can be shaped again, so a modern house can start as one big box: push in the bays, pull out the overhangs. Ctrl+Z undoes a step. Esc cancels.

Shape looks: solid, glass, frosted, glowing

Select a shape you drew (Rectangle, Circle, Line, or one edited with Dots). Under **Look** on the right: **Solid**, **Glass** (clear, see-through), **Frosted** (milky, blurry glass that lets light through) or **Glow** (lit from inside; in the evening it lights the walls and ground around it, and in a photo render it is a real light). **Colour** changes the tint of the glass or the colour of the glow. Material still offers brick, wood, tiles, your own pictures and the rest.

Settings for anything you select: see-through, glow, light

Select something (Ctrl+click several) and press **Settings** in the panel on the right: * **See-through**: 0 to 90 %. * **Look**: Solid, Frosted or Glow. Any object can glow now, not only shapes. * **Light** (for lamps and glowing things): brightness 0 to 300 %, the colour of the light (warm white, white, cool white, red, orange, yellow, green, blue, purple, pink), and which way it shines: all round, down, up, a spot straight down, or aimed (turn and tilt), with the beam width. The light shows in the evening (Sun card) and in photo-real pictures.

New in the library: **Swimming pool** (tiled, coping, a step, clear rippling water), **Pond**, **Frame (square)** and **Frame (flat ring)** that you can size and make glow, and an **LED strip**. Any shape can be water: Material → Water, or the paint bucket's Water.

The tool rail (left)

The tools sit in groups, like SketchUp's toolbars: **Select and draw** (select, wall, line, arc, rectangle, circle, polygon, Dots, opening), **Change** (move, rotate, push/pull, scale, split, offset, Follow Me, array, mirror, paint, group, delete), **Solid tools** (union, subtract, intersect, trim), **Build** (all the Build tools, railing, roof, structure, ceiling, 3D text, add a part, library, import a model), **Measure and view** (tape, measurements on/off, section, hand, zoom, top view, frame, walk, hide, show everything) and **Scene** (site, sun, sky, tiles and stone, the 2D plan). Hover over a button to see what it does.

Wider or narrower: drag the thin strip on the rail's right edge for 1 to 4 buttons across (3 by default, so there is little scrolling). The Layers panel and the right panel can be dragged wider or narrower by their edge strip too (a click on the strip still folds the panel away). Nestify remembers the sizes.

Solid tools: select one shape you drew, Ctrl+click a second one, then Union (one shape), Subtract (the second cut out of the first: a hole, a niche, a pool), Intersect (only where they overlap) or Trim (the first cut out of the second, both kept).

Smart helpers

- **Furnish** (Build group, the house-and-heart button): every named room gets fitting furniture with its back against a wall and clear of the doors: beds and wardrobes in bedrooms, sofa, TV and coffee table in the lounge, a dining table, kitchen counter, stove and fridge, toilet, basin, bath and shower, desk and bookshelf in a study, washer and geyser in the laundry, a car in the garage, and a ceiling light in each room. Rooms are recognised by their names (Lounge, Kitchen, Bedroom ...). Furnish again rearranges only what it put in; "Take out what Furnish put in" removes it. Things you placed yourself are never moved.
- **Running cost:** the status bar shows the building cost of the house as it is now, updated a moment after every change, with how much the last change added or saved. Click it for the full bill of quantities.
- **Describe the picture** (Scene group, the sparkles button): type something like "modern house, golden hour, warm lights, few clouds, winter" and Nestify sets the sun time and month, the sky, the lamps and (for modern, contemporary, farmhouse, tuscan, face brick, stone or wooden) the wall, roof and frame finishes, then offers the photo-real picture.
- **Guided walk video** (Scene group, or Export): click where the walk starts (the front door, or outside it), then click inside each room in the order to visit, and press Enter. Nestify finds the way through the doors by itself, walks at walking pace with the doors open, and stops in every room to look round it (its name and size on screen) before going on to the next: an MP4 in the size you choose.
- **AR page for phones** (Scene group, or Export): one HTML file with the house inside. Send it to a phone, open it in Chrome (Android) or Safari (iPhone) and press "See it in your space": the house stands on a table (1:50) or full size on the real plot. The phone needs to be online.
- **Door or window anywhere** (Select and draw group, the window button): pick a door or window, then click on a drawn wall or on the side of a box: it is cut in right there. In a box it gets a frame (jambs, head and, for a window, a sill) round the door leaf or glass, all grouped together. Click again for more; Esc stops.

Build tools: railings, roofs, ceilings, Follow Me, 3D text

The toolbox button on the left (and the railing, roof, ceiling and sky buttons under it) holds the tools big modelling programs have. Everything they make is an ordinary object: move it, turn it, copy it, paint it, and change every setting later with **Settings** (right panel).

- **Railing**: pick a type (glass panels, vertical bars, horizontal rails, steel cables, timber slats, wrought iron, picket fence or a solid panel), then click point after point where it goes; Enter finishes, clicking the first point takes it all the way round. Settings: height, post spacing, post size, gap between bars, number of rails / cables, handrail (round, square, flat cap, none), bottom rail, square or round posts or frameless glass, and the colours of the frame and of the bars / glass / slats.
- **Roof of your choice**: pick hip, gable, shed (lean-to), flat, gambrel (barn), mansard or pyramid and click the corners of what it covers: a garage, a patio, a carport. Drawn on the ground, it sits on top of the walls. Settings: type, pitch, overhang, thickness and covering (tiles, metal sheets, slate, concrete, timber, glass / polycarbonate). Make as many roofs as you like. The house's

own roof (which follows every wing of the plan) is still there too.

- **Ceilings**: flat, bulkhead (dropped edge), tray (stepped up), LED cove (with a glowing strip in its colour), flat with cornice, coffered (a grid of beams) or sloped; in every room at once, in the selected room, or over any outline you draw.
- **Follow Me**: a profile swept along a path: round pipe, square bar, rectangle, skirting / moulding, kerb or gutter, any width and height, open or as a closed loop.
- **3D text**: words standing up or lying flat, bold or regular, any height and thickness; make them glow in Settings for a sign.
- **Array copies**: many copies in a row (step across, along and up, like 1.2, 0) or round a circle (p 45 for a copy every 45°).
- **Mirror**: flip what is selected left to right or front to back.
- **Offset outline**: a new outline inside (0.1) or outside (+0.1) the top of a shape you drew, ready to Push/Pull: a border, a step, a kerb.

Railings, roofs and pergolas can also be drawn on the **2D plan** (Railing, Roof and Pergola tools): the plan, the colour plan and the printed plans show railings with their posts, and roofs dashed with their ridges, hips and valleys (ceilings show their bulkhead line).

Roof structure and pergolas

Structure (Build group): timber or steel **trusses** with battens / purlins, or **rafters** with battens, under a roof you drew, or under the house's own roof; and **pergola** / **carport** frames (timber or steel posts, beams and joists, open or covered with slats, polycarbonate, metal sheets or shade cloth). In a roof's Settings choose the structure, its spacing, and **Hide the covering** to show the structure. To see the structure under the house's roof, hide the roof (the eye next to Roof in Layers).

Tiles and stone

Material > Tiles and stone (or the Tiles button in the Scene group) on a room's floor, a wall, a shape or the roof: white, grey and black marble, slate, patterned terracotta, white and pool-blue mosaic, crazy paving in sandstone or blue-grey, hexagons, subway tiles, herringbone wood, planks, stacked stone cladding, terrazzo and a black-and-white checker. Each comes with its real size, so the tiles are the right size at once.

Sky

Sky (the cloud button on the left): clear blue, a few clouds, cloudy, overcast, sunset glow or dusk, or **your own sky picture**. A 360 panorama (twice as wide as high) fits best; an ordinary photo of the sky is mirrored round the house so it joins up. The sky shows in the Real look and in reflections.

Shadows and see-through for anything

Settings has **Casts a shadow** / **No shadow** for anything you select, including the lines and outlines

you draw (a Line outline is a shape too: give it glow, see-through or no shadow like any object).

Auto adjust

Ctrl+click the things to tidy (boxes, walls made of boxes, furniture) and press **Auto adjust** in the panel on the right. Within 12 cm it squares them up with the biggest one, puts things that hover or sink just a little right on top of what is under them (or on the ground), closes small gaps and overlaps so faces touch, and lines up edges that are nearly in line. The biggest thing stays where it is; things further apart are left alone. Ctrl+Z puts them back.

Edit group (build one thing on its own)

Like opening a group in SketchUp: double-click a shape, a piece of furniture or a group, or select it and press **Edit group** on the right. The rest of the house turns into faint ghost lines and the camera moves in. Draw rectangles, circles and lines, push / pull them, or add Library items: everything you make joins the group, and only the group's pieces can be clicked and dragged (even behind walls). **Done** (top left) or **Esc** brings the whole house back, at the view you had before.

Your own pictures as materials

Any picture can be a material: floor tiles, marble, wood, brick, stone, wallpaper, a logo. **Material > Your own picture** on a room floor, a wall, the roof or a shape you drew: choose the picture and say how big one picture is in real life. Nestify makes it repeat without seams; it shows in the Real style and in photo-real pictures. A house keeps up to 9 pictures, and they can be reused anywhere.

Which picture works: one tile per picture. Crop the photo to exactly one tile (or one plank, one brick), edge to edge, taken straight on in even light, and type that tile's size (0.6 for a 600 mm tile). A picture of several tiles also works if you type the size the whole picture covers (2 x 2 tiles of 300 mm: 0.6). A photo of a whole room or of tiles at an angle does not work: it repeats as a pattern of photos. Nestify shows this guide before you choose, and warns when a picture looks like many tiles.

Shapes can also be made of **glass** (Material > Glass) - for your own windows, glass boxes and balustrades.

Floors, stairs and rooms

The bar at the top left of the 3D view lists the floors. **+ Floor** adds a storey on top (empty, with the outside walls copied, or a full copy). Click a floor to edit it; the floors above it are hidden so you can work. **All floors** shows the whole house, the roof always sits on the top floor, and ⋮ renames or deletes a floor. Put **Stairs** (Library → Furniture) on a floor and the floor above gets a hole over them; when walking you climb them, or press **PageUp** / **PageDown** to change floor.

Rooms are found automatically from the walls (a doorway counts as closed). Each room shows its name and area in the view (the **Rooms** chip turns the labels off) and in the Layers list, with the total area at the top. Click inside a room to rename it and give it its own floor material and colour; names stay with the room when you move walls.

Walk through the house

Press the walking-person button (or **W**). You start outside the front door at eye height (1.6 m) and walk in like a visitor: **W**/**S** or ↑/↓ walk, **A**/**D** step sideways, ←/→ or **Q**/**E** turn, drag to look around, **Shift** walks faster, the wheel takes a step, and clicking the floor walks you there. Walls, windows and furniture stop you; doors swing open and passages let you through. A mini-map shows where you are (click it to go there) and a **Rooms** list jumps into each room (found automatically, with its area). **Esc** stops walking and puts the camera back where it was.

Structure parts, and lifting objects

The library has a **Structure** tab: round and square concrete columns, brick piers, steel H-columns, timber poles, concrete / steel / timber beams, lintels, roof trusses, purlins, strip footings, slabs, boundary walls, palisade fencing, a carport, a water tank, downpipes and solar panels. Resize any of them like other objects.

To lift anything off the floor (a beam, a shelf, a solar panel onto the roof): **Page Up** / **Page Down** (Shift for 1 cm steps), the **Up 10 cm** / **Down 10 cm** buttons, or **Shift + drag** it up and down. Footings start below the ground; the X-ray view shows them.

Library: furniture, electronics, cars, trees, flowers

The sofa button opens the Library: 37 items (sofas, beds, dining set, wardrobe, kitchen, bathroom, TV, fridge, washing machine, stove, computer, air conditioner, cars, SUV, pickup, trees, pine, palm, bushes, flowers, hedge, fence, bench, pool). Click one, move the mouse to place it and click to drop it. Every object can be dragged (Move), turned (Rotate), resized (Push/Pull or the numbers on the right), recoloured, renamed, copied (Ctrl+D) and saved to **My items** with its name, size and colour. **Import .obj model** adds your own 3D models. My items live in `~/.sketchit/library.json` and `~/.sketchit/models/`.

Lights

The library's **Lights** section has ceiling, pendant, wall and garden fittings: Ring LED (circle), Double ring LED, Round ceiling light, Square LED panel, Pendant (glass globe), Pendant (dome shade), Hanging filament bulb, Chandelier, Track spotlights, square Downlight, Wall light (up and down), Garden bollard, Garden spike spotlight and String lights. Ceiling ones hang from the ceiling of the floor you are on. Settings sets each one's brightness, colour and beam.

Your own 3D models

Library → Import 3D model (or **Build tools → Import a 3D model**) reads **.obj**, **.glb** / **.gltf**, **.dae** (Collada, from SketchUp and Blender) and **.stl**. Each model keeps its parts: one per material, with its colour (and the average colour of its textures). Each part is recognised from its name and colour as leaves, trunk / wood, glass (see-through), a light (it glows and lights the room) or the body; a model

with leaves is filed as a plant. Settings shows every part with its own colour button, plus see-through, glow and light like any other object. Very detailed models (over 60 000 triangles) are simplified on the way in. Compressed glTF (Draco) cannot be read: export it again without compression.

Stairs, railings, site, electrical and plumbing

- **Stairs** (Library > Building): straight, **L-shaped** and **U-shaped** with a landing and a sloping handrail; the floor above gets a stairwell and the walker climbs them, landing included. Also a steel balcony **railing** and a **glass balustrade**.
- **Sloping site**: the mountain button opens the Site card (slope %, the direction it falls, the level platform around the house). The ground is drawn sloping with shadows; **driveway, brick paving, stone path, gravel, instant lawn, timber deck and retaining wall** (Library > Garden) follow it, cars and trees stand on it, and the walker walks on it. The estimate adds paving areas and the cut-and-fill to level the platform.
- **Electrical and plumbing points** (Library > Electrical / Plumbing): plug sockets, switches, TV/data points, outside lights, distribution board and stove isolator snap to the nearest wall at the usual height and face the room; ceiling lights and downlights hang from the ceiling; taps, garden taps, shower, geyser and floor drains. They show as plan symbols (layers ELECTRICAL / PLUMBING in the DXF) and the estimate counts each one plus wiring per electrical point and pipework per wet point.
- **Sun study**: the Sun card films shadows from sunrise to sunset (a frame every 20 minutes, time stamped) as an animated GIF, or MP4 when `imageio[ffmpeg]` is installed.

Ground

The Site card (the hills icon on the left) has a Ground button: grass, dusty soil, gravel and stones, concrete, paving bricks, sand, tar, or **your own picture**. A photo of a lawn, a driveway or a farm is evened out and made seamless, so it repeats without joins; set how wide the area in the photo is. The ground shows in the 3D view, on a sloping plot and in photo-real pictures.

Modern glass and solar panels

Windows: **Glass wall** (floor to ceiling), **Slim frame** and **Clerestory strip**; doors: **Glass slider** and **Glass stacker**. They have slim black aluminium frames split into tall panes, like modern houses. Change the frame colour of any wall's windows and doors (black, charcoal, bronze, silver, white) with **Material > Window and door frame colour** - for one wall or every wall. With a **flat** roof and a deep overhang (Roof settings) and **Natural stone cladding** from the finishes catalogue you get the modern villa look.

Solar panels (Library > Structure) lie on the roof: move one over the roof and it settles on the surface, tilted with the slope and facing down it. Ctrl+D copies it for the next panel.

Real look

The default **Real** style paints plaster, brick, stone or timber walls; wood, tile or concrete floors; tiled, metal or slate roofs; wooden doors and glass windows, under sky light on grass. Pick Material and Colour on the right for a wall, the floor or the roof. Night, Sketch, Neon wire and Blueprint are still there.

Live style pictures

The style pictures on the right turn slowly round the house while you are resting and show exactly your view (with your latest change) while you work. Slow computers get fewer updates; Settings can turn them off.

Pencil and Glass styles

Two more styles on the right: **Pencil** draws the house like a hand sketch - white paper, graphite edges and hatching on the shaded sides. **Glass** makes every face see-through with all its edges showing, so you can see how pieces fit together while you build (it keeps depth, unlike X-ray).

Roof that follows the plan

The roof is built from the real footprint: an L, T or U house gets one wing per arm, the ridges run into each other and meet in valleys, and courtyards stay open. Hip, gable, shed, flat, gambrel and mansard all work this way; slanted outlines get a true hip over the outline. Eaves overhang is measured from the outside face of the walls. While the Wall tool is active the roof steps aside so you can see the plan.

Sun and shadows

The sun button on the left opens the Sun card: time of day, month, place (city list), and which way north points on the plan. The house casts real shadows on the ground, on itself and from trees and cars, and the light turns warm in the morning and evening. **Shadows** and **Real sun** can be switched off.

X-ray (engineering view)

The X-ray card in Styles turns the house into a see-through engineering model: glowing blue surfaces, every edge drawn (hidden ones too) and a mesh of lines across walls, floors and furniture. Click any wall, object or anything you built and it glows blue while everything else turns into grey ghost lines, so you can study that one part inside the whole house. Click empty space to see everything in blue again.

Photo-real pictures

Press the camera-shutter icon (top right) or Export → Photo-real picture. Nestify hands the view you

see to a physically based path tracer (Mitsuba 3): a real sun and sky for your place, day and hour; light that bounces around the rooms and comes in through the windows; glass that reflects; plaster, face brick, stone, timber floors, tiles and roof tiles as real materials. Choose:

- **Size**: Small, HD (1280 × 720), Full HD, Square.
- **Quality**: Quick (a draft in seconds), Good, Best (for printing).
- **Light**: as the Sun card, Morning, Golden hour, Evening with the lights on, Night.

Photos need a processor with AVX instructions (almost every Core i3/i5/i7 or Ryzen PC; not low-power Celeron, Pentium Silver or Atom chips). On those PCs the Photo button says so and offers Picture of this view.

The picture sharpens pass by pass while you watch, is cleaned of grain, toned like a photo, and kept in Pictures/Nestify as well as wherever you save it. Walk inside (W) for a photo of a room: from inside, the windows are open to the sun and the sky.

Lamps (ceiling lights, downlights, floor lamps, outside wall lights) glow, and in the evening (or with Lights on in the Sun card) they light their own rooms in the 3D view and are real lights in photos. **Curtains** (open, closed, and net curtains that let light through) are in the library under Furniture; generated houses get curtains on their windows. **TVs** (on a stand, on the wall, a computer screen) can show a picture or play an animated GIF: select the TV and press "Picture / GIF on the screen". It plays in the 3D view and shows in photos.

Building rules check

The clipboard button in the top bar checks the house against SANS 10400 rules of thumb: habitable rooms at least 6 m² and 2 m wide, ceilings 2.4 m (2.1 m for bathrooms and passages), windows about 10% of the floor for daylight and 5% opening for fresh air, doors 750 mm, stair risers up to 200 mm and treads 250 mm, outside walls 140 mm or more, stairs to every upper floor. Tap a finding to go to it. A first check, not an approval.

Quantities and cost

The calculator button in the top bar opens a bill of quantities worked out from the model: strip footings (600 x 230) and floor slabs in concrete, bricks (52 per m² per leaf, +5%), cement and sand for mortar and plaster, paint (2 coats), floor tiles / wood / screed per room, ceiling boards, roof tiles or sheets from the real roof surface, roof timber, gutters, and every door and window. Each line is priced; labour and contingency are added as percentages to get the total and a rate per m². **Prices** edits every rate (saved in `~/.sketchit/prices.json`). The currency follows the computer's country (Windows region setting): rand in South Africa, naira in Nigeria, pounds in the UK and so on, with the built-in prices converted at today's exchange rates as a starting point. Rates are fetched in the background from open.er-api.com (free, no account) when the app starts, kept in `~/.sketchit/rates.json` and refreshed every 12 hours; without internet the last rates are used (**Update rates** in Prices fetches them now); pick another currency at the top of Prices, either converting the numbers or only changing the symbol; **PDF** and **CSV** / **Excel** save the estimate with a room schedule. Rules of thumb for planning, not a quantity surveyor's bill.

Tape measure

E (or the tape button) measures between two clicked points, snapping like the Wall tool. Kept sizes show in the 3D view and print on the 2D plan; Ctrl+Z removes the last, the floor menu clears them.

Plans, council pack and pictures

Export (toolbar) offers:

- **2D plan PDF**: one A3 page per floor at a standard scale (1:50, 1:100 ...), with walls, doors and their swing, windows, stairs, furniture, room names and areas, dimension chains, scale bar, north arrow and a title block. Sizes are in the unit chosen at the bottom right.
- **2D plan PNG** of the floor you are on.
- **Elevations and sections**: the plan PDF continues with the four elevations (named by the compass from the north in the Sun card) and sections A-A and B-B drawn from the 3D model: clean lines, pale-blue glass, timber doors, grey roof, cut walls / slabs / roof in solid black, level marks (FFL of every floor, wall plate, ridge) and the overall height. The section lines are marked on every floor plan. "Elevations and sections only" saves just those pages.
- **DXF** for AutoCAD, LibreCAD, SketchUp and other CAD programs (layers WALLS, DOORS, WINDOWS, ROOMS, DIMS, FURNITURE; floors side by side, millimetres).
- **Picture of this view** at 2x or 3x the window size, without grips, labels or selection.
- The 3D model as OBJ or STL, as before (all floors and the roof).
- **Council submission pack** (PDF): site plan (plot boundary, building lines, street, roof outline, driveway, setback dimensions, coverage), every floor plan, elevations, sections, a 1:25 wall section (footing, DPC, slab, wall plate, roof build-up), and schedules (rooms and areas, doors, windows, checks, general notes). Every sheet carries the owner, erf and address from **Project details**, which the pack asks for the first time.
- **Share as a web page**: one .html file with the house in 3D inside it. Anyone can open it in a browser, on a phone too, with no app or internet: turn it with a finger or the mouse, Top / Front views, take the roof off, hide the upper floors, and see the room list.

Colour plan for clients

Export → **Colour plan for clients** draws the floor plan the way plan books and estate agents show it: thick red walls (or black, grey or blue), light tiled floors with small tiles in bathrooms, kitchen and laundry, paving in the garage and on patios, windows as pale blue glass, doors with their swing shaded timber, every piece of furniture, bathroom fitting and car seen from above in its own colours with a soft shadow, and each room's name in bold capitals with its area. Names move to a free spot so furniture never covers them. It is cut at 1.2 m like an architect's plan, so houses built from boxes work too. PNG for this floor, or a PDF with every floor.

Sides of the house (your own views in the plans)

Turn the 3D view to a side you like and press **Sides** (the bar over the 3D view) → **Save this view as a side**: Front, Back, Pool side ... **Plans to print** then prints your sides (four to an A3 page, named, not to scale) instead of the drawn elevations; the sections stay. Choose how they look on paper (Real, Sketch, Pencil, Blueprint, Glass) in the same menu. Update, rename or delete a side there too. With no sides saved, the drawn elevations print as before.

Section view

The **Section** chip above the 3D view cuts the house open like a builder's section drawing. **Cut across** slices it at a height (1.2 m above the floor to start) so you look down into the rooms; **Front to back** and **Side to side** slice it upright, cutting away the half nearest you. Drag the slider to move the cut, and **Flip** to look from the other side. Cut walls show solid red, the way sections are drawn. Pictures you save keep the cut; **Off** (or the Section chip again) closes it.

Kitchen builder

Select the wall the kitchen goes against and press **Fit a kitchen on this wall** (on the right). Pick what you want: cupboards, a sink, a stove with an extractor hood and a fridge; without the fridge; with a tall pantry as well; or base cupboards only. Nestify uses the longest free stretch of the wall (doors and low windows are kept clear, up to 5.4 m), sizes the cupboards to fill it exactly, and leaves wall cupboards out under windows. The kitchen is one group: drag it to move it all, or Ungroup to change single units. The pieces are also in the **Kitchen** section of the Library.

Finishes catalogue

Select a room floor, a wall or the roof and press **Material**: at the top is **Choose a product**, a list of real finishes with their prices: floor tiles, porcelain, laminate, vinyl plank, solid wood, polished concrete, carpet tiles; paint colours, face brick, stone and timber cladding; concrete and clay roof tiles, IBR, corrugated and Klip-lok sheeting, slate. The 3D view takes the product's look, and **Quantities and cost** prices it on its own line (with waste). Prices follow your price list's currency. Add your own products in `~/.sketchit/materials.json`.

360 pictures and walkthrough videos

Export > 360 picture of a room: pick a room (or where you stand in walk mode) and Nestify saves a 360 photo from eye height. Post it on Facebook or open it on a phone and turn around in it.

Export > Walkthrough video makes an MP4: a slow flight around the house, a glide through your saved Scenes, then a look around every room with its name and size. Choose standard, HD or full HD.

Export the 3D model

Export > 3D model in colour: a **.GLB** (glTF) file opens in Blender, SketchUp (glTF extension),

Windows 3D Viewer and 3D web viewers; **.OBJ** comes with a **.MTL** colour file beside it. Both include the roof, doors, windows and furniture, in metres. STL (for 3D printing) is still there.

Updates

Once a day Nestify checks the developer's website for a newer version and offers the download (Help > Updates checks now). Copies from the Microsoft Store are updated by the Store instead.

Files

Save writes a `.json` project (atomically). Unsaved work is autosaved every 45 s and offered back after a crash. Export writes the 2D plan (PDF, PNG, DXF), a picture of the view, or the 3D model as OBJ (SketchUp, Blender) or STL (3D printing), in millimetres. Floors, rooms and the sun are saved in the project; older projects open as one floor.

Working with big files

The Layers list is updated in place instead of rebuilt (it used to take ~0.4 s per click), the roof is not rebuilt on every mouse move while dragging, and big changes (roof type, new plan, scaling, settings) are built in the background behind a "please wait" card.